Dongwon Kim

@ kdwon@postech.ac.kr | https://kdwonn.github.io

Research summary

Research scientist working on world models and embodied AI, with a focus on representation learning and generative modeling that let world models learn fast and plan cheaply. Author of five first-author papers at CVPR/ICCV/NeurIPS (incl. a CVPR Highlight), with research experience at ByteDance Seed on scalable text-to-image generation.

Education

POSTECH — Pohang, South Korea
Integrated M.S. and Ph.D. in Computer Science and Engineering; — *Sep 2019 – Aug 2025*

- Supervised by Prof. Suha Kwak in the Computer Vision Lab.
- Thesis title: Learning Compositional Visual Representations for Vision-Language Understanding and Generation
- CSE Best Research Award — best doctoral dissertation among the department's 2025 PhD graduates
- Research interest: Computer vision, multi-modal learning, representation learning, generative modeling

POSTECH — Pohang, South Korea
B.S. in Computer Science and Engineering — *Mar 2015 – Aug 2019*

Work experience

KAIST — Daejeon, South Korea
Postdoctoral Researcher – InnoCore-LLM Research Center — *Nov 2025 – Present*

- Researching efficient world models and planning with compact visual representations

ByteDance SEED — San Jose, California
Research Intern – Fundamental Research Team — *June 2024 – Nov 2024*

- Developed efficient text-to-image generative model using compact text-aware 1D tokens
- Project published as "Democratizing Text-to-Image Masked Generative Models..." (accepted to ICCV 2025)
- Worked with Ju He, Qihang Yu, and Liang-Chieh (Jay) Chen

Publications

[1] ***Reliable Action-Chunk Extension for Vision-Language-Action Models*** | arXiv
Seonghoon Yu, **Dongwon Kim**, HyungRok Jung, Yoonjae Baek, Byung-kwan Lee, Suha Kwak, Jeany Son
arXiv preprint, Oct 2026

[2] ***ACID: Action Consistency via Inverse Dynamics for Planning with World Models*** | arXiv
Gawon Seo, **Dongwon Kim**†, and Suha Kwak†
arXiv preprint, Jul 2026

[3] ***Planning in 8 Tokens: A Compact Discrete Tokenizer for Latent World Model*** | arXiv
Dongwon Kim, Gawon Seo, Jinsung Lee, Minsu Cho, and Suha Kwak
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Jun 2026

[4] ***Structured State-Space Regularization for Compact and Generation-Friendly Image Tokenization*** | arXiv
Jinsung Lee, Jaemin Oh, Namhun Kim, **Dongwon Kim**, Byung-Jun Yoon, Suha Kwak
arXiv preprint, Apr 2026

[5] ***Democratizing Text-to-Image Masked Generative Models with Compact Text-Aware One-Dimensional Tokens*** | arXiv
Dongwon Kim*, Ju He*, Qihang Yu*, Chenglin Yang, Xiaohui Shen, Suha Kwak, and Liang-Chieh Chen
IEEE/CVF International Conference on Computer Vision (ICCV), Oct 2025

[6] ***1.58-bit FLUX*** | arXiv
Chenglin Yang, Celong Liu, Xueqing Deng, **Dongwon Kim**, Xing Mei, Xiaohui Shen, Liang-Chieh Chen
arXiv preprint, Dec 2024

[7] ***Bootstrapping Top-down Information for Self-modulating Slot Attention*** | arXiv
Dongwon Kim, Seoyeon Kim, Suha Kwak
Neural Information Processing Systems (NeurIPS), Dec 2024

[8] ***Text-based Person Search with Part Slot Attention for Corresponding Part Discovery*** | arXiv
Jicheol Park, **Dongwon Kim**, Boseung Jeong, Suha Kwak
European Conference on Computer Vision (ECCV), Oct 2024

[9] ***Extending CLIP's Image-Text Alignment to Referring Image Segmentation*** | arXiv
Seoyeon Kim, Minguk Kang, **Dongwon Kim**, Jaesik Park, Suha Kwak
Annual Conference on the North American Chapter of the Association for Computational Linguistics (NAACL), Jun 2024

[10] ***Shatter and Gather: Learning Referring Image Segmentation with Text Supervision*** | arXiv
Dongwon Kim*, Namyup Kim*, Cuiling Lan, Suha Kwak
IEEE/CVF International Conference on Computer Vision (ICCV), Oct 2023

[11] ***Improving Cross-Modal Retrieval With Set of Diverse Embeddings*** | arXiv
Dongwon Kim, Namyup Kim, Suha Kwak
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Jun 2023
Highlight (Top 2.5% = 235/9155)

[12] ***ReSTR: Convolution-Free Referring Image Segmentation Using Transformers*** | arXiv
Namyup Kim, **Dongwon Kim**, Cuiling Lan, Wenjun Zeng, Suha Kwak
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Jun 2022

[13] ***Self-Taught Metric Learning Without Labels*** | arXiv
Sungyeon Kim, **Dongwon Kim**, Minsu Cho, Suha Kwak
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Jun 2022

[14] ***Embedding Transfer With Label Relaxation for Improved Metric Learning*** | arXiv
Sungyeon Kim, **Dongwon Kim**, Minsu Cho, Suha Kwak
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Jun 2021

[15] ***Proxy Anchor Loss for Deep Metric Learning*** | arXiv
Sungyeon Kim, **Dongwon Kim**, Minsu Cho, Suha Kwak
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Jun 2020

Awards & Achievements

POSTECH CSE Best Research Award, POSTECH, 2026

- Awarded for the best doctoral research among 2025 PhD graduates in the CSE department.

Best Paper Honorable Mention, IPIU, 2026

Postechian fellowship, POSTECH, 2023

BK21 Best Paper Award, POSTECH GSAI, 2023

Qualcomm Innovation Fellowship Winner, Qualcomm Korea Corp., 2022

NAVER × POSTECH AI Day, 2nd and 3rd Prize, NAVER, 2022

Qualcomm Innovation Fellowship Winner, Qualcomm Korea Corp., 2021

Best Paper Award, IPIU, 2021

National Science & Technology Scholarship, Korea Student Aid Foundation, 2017-2018

Jigok Scholarship, POSTECH, 2015-2016